A Sensor-Aware Hypothesis Bank for Passive Frequency-Hopping Generator Identification, Next-Channel Prediction and Calibrated Response

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***Abstract*—A passive electronic-support receiver that watches a frequency-hopping (FH) emitter must decide, inside a short engagement window and through a lossy sensor, which generator drives the hop sequence, what channel comes next, and whether the stream is predictable at all. We present OTS, a software system that scores a heterogeneous bank of generator hypotheses—shift registers, congruential and weak-seed generators, chaotic maps, a Markov hopper, and an explicit generative none-of-the-above class—through one shared detection-level likelihood parameterised by the sensor's detection and false-alarm probabilities. The same likelihood drives anytime-valid stopping, a training-free next-channel predictor that keeps working on generators the bank has no expert for, a change-point tracker that detects and re-identifies a mid-engagement change of generator, and a conformal prediction set whose size adapts to give a distribution-free coverage guarantee that transfers directly to the fraction of hops a multi-tone response denies. Across more than $15{,}000$ simulated episodes spanning thirteen generators and four sensor operating points, in-bank families are identified at ≥ 0.99 accuracy in 4–13 hops, the dangerous error of falsely naming an algebraic generator stays at or below 0.05 everywhere, next-channel prediction reaches 1.00 steady-state top-1 on held-out generators where an identification-only bank collapses to 0.35, and conformal sets hold ≥ 0.90 coverage where a fixed shortlist covers as little as 0.19. The measured hops-to-identify sits within 1.3–$2.7\times$ an information-theoretic lower bound.**

***Index Terms*—frequency hopping, electronic support, sequential hypothesis testing, online prediction, conformal prediction, change-point detection.**

I. Introduction

Frequency hopping (FH) is the standard way a radio link resists interception and jamming: the carrier moves between channels under a pseudo-random generator known only to the communicating parties. A passive electronic-support (ES) receiver that wishes to characterise such an emitter faces three coupled questions. *Which* generator is in use, so that past hops constrain future ones? *What* channel will be used next, which is the quantity any response must act on? And *how predictable* is the emitter at all, since a keyed cryptographic generator is, by design, indistinguishable from noise and should be reported as such rather than chased. These questions must be answered inside an engagement window of tens of hops and through a sensor that misses true hops and reports phantom ones.

Most prior work answers only one question and assumes a clean stream. Neural predictors forecast the next hop of a specific family from an idealised sequence [1]–[4]; sequential detectors identify a parameter but not a heterogeneous family set [8], [9]; particle filters track a hopped carrier but not the *algorithm* behind it [13], [14]. None scores a bank of competing generator families through an explicit sensor model, and none couples a calibrated forecast to a decision with a stated guarantee.

This paper presents OTS, a system built around a single idea: reduce the lossy sensor to a per-slot *detection-level likelihood* and use that one quantity everywhere. The contributions, each backed by a simulation experiment under the information-parity rule that every expert sees only the sensor's detections, are:

1) A unified sensor-aware hypothesis bank that scores shift-register, congruential, weak-seed, chaotic and Markov generators plus an explicit generative none-of-the-above class H_0 through one (P_d, P_{fa}) likelihood, with MSPRT and anytime-valid e-process stopping (Sec. III-B).
2) A training-free, online next-channel predictor (a Fixed-Share mixture carrying a universal context-mixing expert) that keeps predicting on generators the bank has no expert for, and correctly falls to chance on a cryptographic generator (Sec. III-C).
3) A change-point tracker that detects, dates and re-identifies a mid-engagement change of the generating algorithm from the lossy passive stream (Sec. III-D).
4) A conformal next-channel set whose size adapts to give a distribution-free coverage guarantee, which transfers to a "fraction-of-hops-denied" guarantee for a multi-tone response (Sec. III-E).

We evaluate on thirteen generators across four sensor operating points, report a mis-specified-sensor stress test, and compare the measured identification cost against an information-theoretic converse.

II. Related Work

FH prediction. Deep models predict FH sequences from time–frequency representations [1]–[3], [7], including CNN–

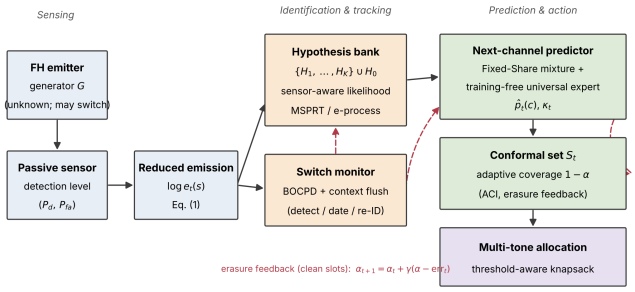


Fig. 1. System architecture. One detection-level likelihood feeds the hypothesis bank, the switch monitor, the next-channel predictor and the conformal set; the set drives the multi-tone response. Dashed red: the switch monitor resets the bank/predictor, and erasure feedback calibrates the conformal level online.

LSTM drone FHSS occupancy validated on real signals [4]; all are trained per family on clean spectrograms. The closest precedent to our "identify, predict, flag unknown" loop reverse-engineers LFSR hoppers with a neural network [5], but is LFSR-only, trained, and clean-data; transformers predict linear-congruential outputs in-context [6]. Our predictor differs by being training-free, online, scored under detection losses, and tied to an identification bank.

Sequential tests and sensor likelihoods. Threshold stopping is the MSPRT of [8], optimal under correlation [9]; the e-process is a safe anytime-valid test [10]; the generative none-of-the-above class is the no-match idea of [11]. The reduced emission is the discrete-channel detection/clutter likelihood behind probabilistic data association [12]; particle filters track a hopped carrier [13], [14] but not the algorithm.

Change-point, conformal and application. A run-length posterior carrying candidate models is the architecture of [15], built on [16]; our set uses Adaptive Conformal Inference [17] for non-exchangeable streams; recurrence recovery draws on classical cryptanalysis [18], [19]. Bluetooth hop-prediction attacks [20] are a deployed intercept-and-act instance, and learned anti-jamming hopping is the dual problem [21]; we treat the multi-tone response only as a downstream use of the calibrated forecast, not a new jammer.

III. System Model and Methodology

Fig. 1 shows the end-to-end data flow and Fig. 2 the per-slot loop (Algorithm 1).

A. Problem and sensor model

A single emitter transmits, at slot $t = 1, \dots, T$, on channel c_t from a set C of size M (default $M = 16$), chosen by an unknown generator G. A passive sensor detects the true channel with probability P_d and raises an independent false alarm on every other channel with probability P_{fa}, producing a detection set $D_t \subseteq C$. For a hypothesised true channel s, factoring out the all-false-alarm constant F_t (identical across hypotheses) leaves the *reduced emission*

$$\log e_t(s) = \begin{cases} \log(P_d/P_{fa}), & s \in D_t, \\ \log\big((1-P_d)/(1-P_{fa})\big), & s \notin D_t. \end{cases} \quad (1)$$

F_t cancels in every posterior and likelihood ratio, so the whole system works with (1), which is used twice: once to generate observations and once inside every expert. Because the same simulator generates adversaries and scores them, we enforce *information parity*: an expert sees only the stream of observations and its own bounded reconnaissance ranges, never the true channel sequence or the generator instance.

B. The hypothesis bank and stopping rules

The bank keeps a posterior over $\{H_0, H_1, \dots, H_K\}$ and updates it after every slot with each expert's reduced likelihood, $\pi_t(k) \propto \pi_{t-1}(k)\, P(o_t \mid o_{1:t-1}, H_k)$, with a floor on the H_0 prior so the unknown class is never ruled out. Deterministic families (LFSR, filtered LFSR, LCG, weak-seed) are realised by a particle filter over the family's parameter space: each candidate parameterisation is one particle whose deterministic trajectory is scored through (1), which is immune to gaps because it never needs consecutive outputs. Markov and chaotic families use a channel-level HMM forward pass.

Two stopping rules read the cumulative reduced log-likelihoods. *MSPRT* declares $k^* = \arg\max_k \pi_t(k)$ when $\pi_t(k^*) \geq 1 - \delta$; $k^* = 0$ means *unknown*. The *e-process* declares k when the e-value against its strongest competitor, $E_t(k) = P(o_{1:t} \mid H_k) / \max_{j \neq k} P(o_{1:t} \mid H_j)$, crosses $1/\beta$; by Ville's inequality this bounds the probability of ever declaring a wrong in-bank k by β per competitor.

C. Next-channel prediction

Identification is a means; the deliverable is the next channel. A Fixed-Share expert mixture weights experts by their *recent* one-step predictive skill,

$$w_{t+1}(i) \propto w_t(i)\, e^{\eta\, \ell_t(i)}, \quad w_{t+1} \leftarrow (1-\alpha) w_{t+1} + \alpha/N, \quad (2)$$

and mixes their forecasts, $\hat{p}_t(c) = \sum_i w_t(i)\, p_i(c)$. The share floor α lets a dormant expert recover after a change. Crucially, the mixture always carries a *training-free universal* expert—a gap-robust variable-order context-mixing model—so when no identification expert fits a new generator, the universal expert accrues weight and carries the forecast instead of collapsing to the uniform floor. The system also reports the predictability score $\kappa_t = 1 - H(\hat{p}_t)/\log M$.

D. Switch tracking

A cumulative-evidence identifier is slow to forget. A Bayesian online change-point posterior whose run-length hypotheses each carry a bank *anchored at the hypothesised switch slot* lets a freshly re-keyed or counter-based generator restart at zero, so one posterior detects, dates, re-identifies and predicts simultaneously. A cheaper CUSUM monitor on the mixture's loss stream additionally resets the Fixed-Share weights and flushes the universal predictor's stale context on an alarm.

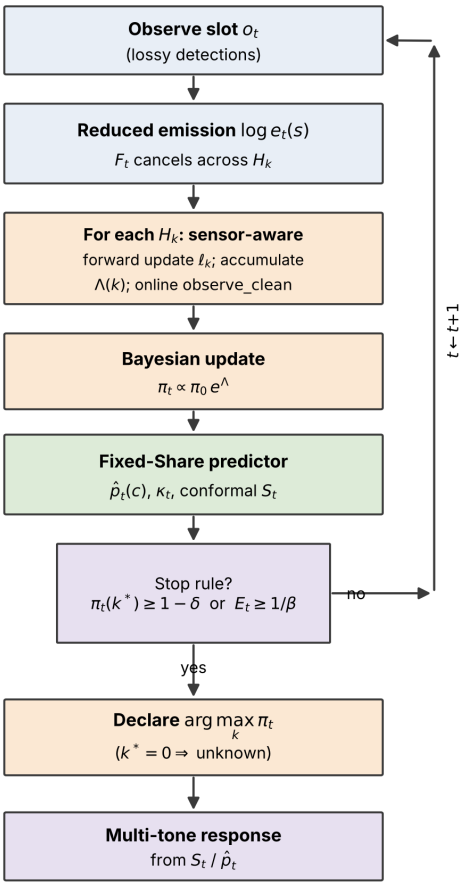


Fig. 2. Per-slot engagement loop (Algorithm 1). One reduced emission drives the Bayesian update, the Fixed-Share predictor and the stopping test; the loop runs until a declaration or the window ends.

E. Conformal sets and response

A point forecast carries no guarantee. We emit a prediction set S_t: the smallest top-ranked set whose predictive mass reaches an online level $1-\alpha_t$, with α_t updated by Adaptive Conformal Inference [17], $\alpha_{t+1} = \mathrm{clip}(\alpha_t + \gamma(\alpha - \mathrm{err}_t), 0, 1)$. The calibration signal err_t is the erasure feedback a passive node actually receives (clean slots only), so coverage holds over the clean sub-stream without assuming exchangeability. Because the set *is* the multi-tone jam plan, funding every channel in S_t denies the realised next hop at the coverage rate, so the fraction of hops denied inherits the distribution-free $1-\alpha$ bound. A threshold-aware greedy knapsack (rank channels by predicted mass per unit denial cost, fund to a budget) is the underlying allocator.

IV. Experimental Setup

All experiments are simulation-only with $M = 16$, fixed seeds, and figures regenerated from saved JSON. Table I lists the generator library. Three "held-out" families (lagged Fibonacci, quadratic congruential, sine map) have *no* dedicated expert and test prediction generalisation; a keyed SHA-256 CSPRNG is the unpredictable control. We sweep four sensor operating points: Perfect $(1.0, 0)$, Strong $(0.95, 0.01)$, Typical $(0.8, 0.02)$ and Contested $(0.6, 0.05)$. Identification uses MSPRT with $\delta = 0.05$. We report identification accuracy, hops-to-identify τ, and the *hard* false-ID rate—declaring an algebraic family (LFSR, filtered LFSR, LCG, weak-seed) that is not the truth, the dangerous error. Prediction is scored by steady-state top-1 next-channel accuracy against the chance line $1/M = 0.0625$. The headline identification evaluation aggregates 100 seeds $\times$ 13 generators $\times$ 4 sensors (5,200 episodes); the coverage experiment adds 7,500.

Algorithm 1 Engagement loop (one emitter, window T)

$\pi \leftarrow \pi_0$; reset every expert
for $t = 1$ **to** T **do**
  $o_t \leftarrow \mathrm{observe}()$; form $\log e_t$ by (1)
  for $k = 0$ **to** K **do**
    $\ell_k \leftarrow H_k.\mathrm{step}(\log e_t)$; $\Lambda(k)$ += ℓ_k; $H_k.\mathrm{observe_clean}(o_t)$
  end for
  $\pi \leftarrow \mathrm{softmax}(\log \pi_0 + \Lambda)$
  $\hat{p} \leftarrow$ Fixed-Share predictor; form set S_t
  if stop rule fires **then**
    return declare$(\arg\max_k \pi)$, $\tau = t$
  end if
end for
return declare$(\arg\max_k \pi)$, $\tau = T$

TABLE I
Generator library. Held-out families have no expert.

Family	Structure	Hidden parameters
LFSR / filt. LFSR	shift register (+ combiner)	taps, length, state
LCG / full LCG	$x \leftarrow (ax+c) \bmod m$	m, a, c, seed
weak-seed	small-keyspace PRNG	seed
Markov	first-order hopper	transition matrix
logistic/tent/Hénon	chaotic maps, quantised	map parameter, x_0
lagged-Fib. (held out)	$c_t = (c_{t-1} + c_{t-k}) \bmod M$	order k
quad.-CG (held out)	$x \leftarrow (ax^2 + bx + c) \bmod M$	a, b, c
sine map (held out)	$x \leftarrow r \sin(\pi x)$	r, x_0
CSPRNG (control)	SHA-256 counter mode	key

V. Results

A. Identification: coverage and safety

Fig. 3 shows family confusion matrices at the four sensors. The diagonal is clean for every generator an expert targets. As the sensor degrades, errors are *graceful*: a held-out generator or a worn-down Markov source routes to the Markov or H_0 column, never to a false algebraic claim. Table II aggregates the identifiable families. Deterministic families are named at 1.00 accuracy in a handful of hops even on the contested sensor; the hard false-ID rate stays at or below 0.05 throughout, and is 0.00 for the shift-register and weak-seed families that an operator would most dangerously misattribute.

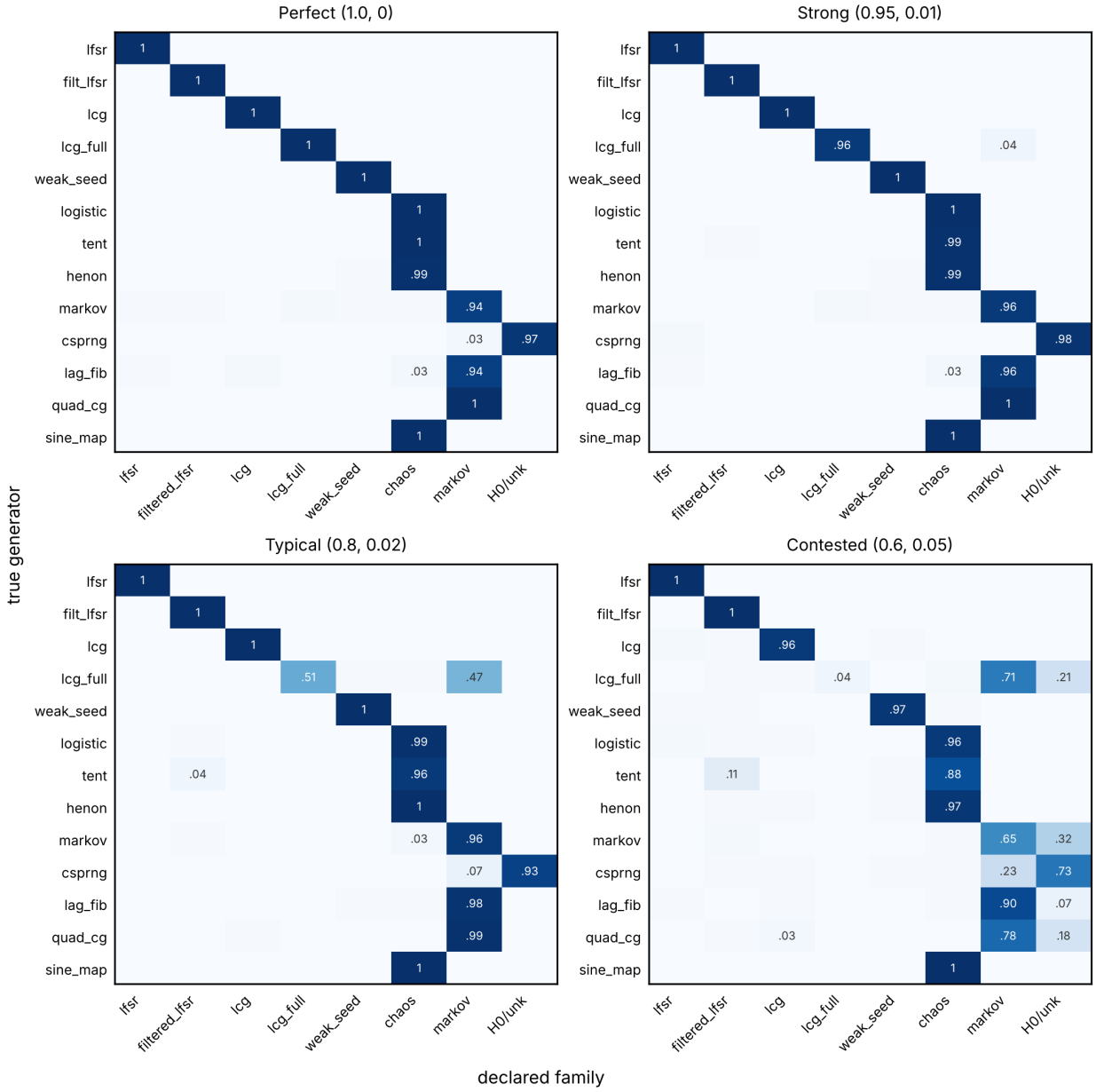


Fig. 3. Family confusion matrices (true generator vs declared family), one per sensor, 100 seeds per cell. Rows are the thirteen generators including the three held-out families and the CSPRNG; columns are the declared families and H_0/unknown. The diagonal stays clean for identifiable families; held-out and unbreakable sources degrade to Markov/H_0, never to a false algebraic claim, even on the contested sensor.

B. The value of the sensor-aware likelihood

Replacing (1) with a naive clean-match score leaves the two in agreement while detections are plentiful, but as P_d falls to 0.5 (at $P_{fa} = 0.05$) the naive bank drops to 0.60 accuracy against 0.98 and needs 21.1 hops against 15.3, because it reads a missed hop as evidence against the truth and a phantom as evidence for a wrong generator. At the level of a single recovery algorithm the effect is a cliff: plain Berlekamp–Massey recovers an L=7 LFSR on every run at P_d=1 and on *no* run for any $P_d < 1$, while the state-enumeration recovery under (1) holds 1.00 across the whole sweep—the gap is exactly the value of modelling the sensor inside the likelihood.

C. Mis-specified sensor

The bank scores with the same (P_d, P_{fa}) that generated the data; a referee will ask what happens when they differ. Fig. 4 sweeps the assumed value against the true one. Identification is strikingly robust to an over-optimistic P_d: accuracy stays high across the off-diagonal, because a missed hop merely omits a likelihood factor rather than corrupting one, and the hard false-ID rate under a mis-specified P_{fa} stays low. The system is sensor-*aware* rather than sensor-*oracle*: it degrades gently rather than failing when its sensor estimate is wrong.

TABLE II
IDENTIFICATION ACCURACY / HARD FALSE-ID ACROSS SENSORS (100 SEEDS).

Generator	Perfect	Strong	Typical	Contested
	accuracy / hard false-ID			
LFSR	1.00 / .00	1.00 / .00	1.00 / .00	1.00 / .00
filt. LFSR	1.00 / .00	1.00 / .00	1.00 / .00	1.00 / .00
LCG	1.00 / .00	1.00 / .00	1.00 / .00	0.96 / .04
weak-seed	1.00 / .00	1.00 / .00	1.00 / .00	0.97 / .00
chaos (log.)	1.00 / .00	1.00 / .00	0.99 / .00	0.96 / .00
Markov	0.94 / .05	0.96 / .01	0.96 / .01	0.65 / .02
CSPRNG	0.97 / .00	0.98 / .00	0.93 / .00	0.73 / .03

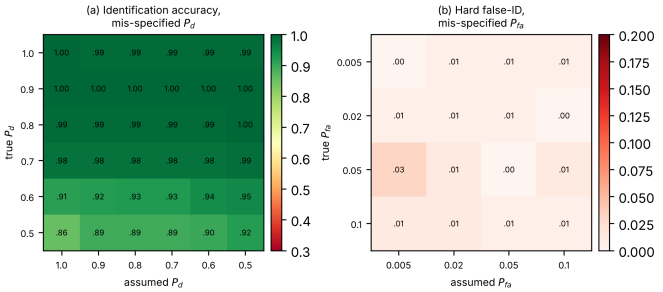


Fig. 4. Mis-specified sensor. (a) Identification accuracy as the assumed P_d (columns) departs from the true P_d (rows). (b) Hard false-ID as the assumed P_{fa} departs from the true P_{fa}. Accuracy and safety hold across a wide mismatch band.

D. Next-channel prediction and conformal sets

Fig. 5 collects the headline prediction results. Panel (b) is the central claim: on the held-out lagged-Fibonacci generator the proposed predictor reaches 1.00 steady-state top-1 while an identification-only bank manages 0.35, because the universal context-mixing expert recovers the order-2 rule that no bank expert can represent. On a keyed CSPRNG both correctly fall to chance. Table III gives the operating-point numbers; the advantage holds on the typical and contested sensors. Panel (c) shows the conformal set tracking the target coverage diagonal while a fixed top-3 shortlist sits well below it—a fixed shortlist covers only 0.51 on a Markov hopper and 0.19 on a CSPRNG, whereas the conformal set holds ≥ 0.90 by growing its size (one tone on a pinned-down LFSR, up to the whole plan on the CSPRNG), so the number of transmit frequencies is set by the data.

E. Switch tracking and the cost of identification

Table IV replays 24 random switch plans over all thirteen algorithms (113 switches) through the switch tracker. Switches are detected in 95–97% of cases with a median delay of 0 hops on a perfect sensor and 12 on a contested one; dangerous locks stay at or below 0.015 per regime (two across the whole contested run). Finally, the measured hops-to-identify sits within 1.3–$2.7\times$ an information-theoretic lower bound $\mathbb{E}[\tau] \geq \log((1-\delta)/\delta)/D$ with the posterior log-odds drift D estimated on the runs (mean efficiency $2.2\times$, always at or above the floor), so the empirical stopping law is near-tight rather than merely a good fit.

TABLE III
STEADY-STATE TOP-1 NEXT-CHANNEL ACCURACY, PROPOSED PREDICTOR (P) VS IDENTIFICATION-ONLY BANK (I), ACROSS SENSORS (CHANCE $= 0.0625$, 30 SEEDS).

Generator	Strong		Typical		Contested	
	P	I	P	I	P	I
LFSR (in-bank)	1.00	1.00	1.00	1.00	1.00	1.00
lagged-Fib. (held out)	**1.00**	0.35	**0.94**	0.33	**0.43**	0.16
quad.-CG (held out)	1.00	1.00	**0.98**	0.50	**0.68**	0.09
sine map (held out)	0.44	0.44	0.38	0.38	0.27	0.26
CSPRNG (control)	0.07	0.07	0.07	0.06	0.06	0.06

TABLE IV
SWITCH TRACKING: 24 RANDOM PLANS OVER 13 ALGORITHMS (113 SWITCHES), $M = 16$.

Sensor	Perfect	Strong	Typical	Contested
Switches detected	97%	97%	97%	95%
Delay, median/90th (hops)	0/6	1/7	4/12	12/34
Dangerous locks /regime	.000	.000	.000	.015
Top-1, hops 6–10	0.72	0.65	0.53	0.20

VI. DISCUSSION

One likelihood explains the curves. Every deterministic family is scored by enumerating its parameters through (1). This single choice accounts for single-digit hops to identify, cost that scales as $1/P_d$, and immunity to false alarms—a phantom scales every candidate equally and cancels.

Hard versus soft errors. The dangerous error is a confident, false claim of an algebraic break; it would send an operator after the wrong generator. That rate is at most 0.05 across every sensor and all coverage episodes, while a closed-set test without H_0 makes it on every out-of-bank source. The case for an explicit generative H_0 is this asymmetry, not raw accuracy.

A limited bank does not limit prediction as much. Any bank is finite, and an adversary can pick a generator outside it. Because the Fixed-Share mixture weights experts by recent skill, the training-free universal expert takes over within a few slots; the lagged-Fibonacci result (1.00 vs 0.35) is the clear case. Where a source is only statistically structured (sine map, Markov), the mixture does no better than its best statistical learner—the honest limit.

Limitations. All results are simulated with a slot-synchronous, i.i.d. sensor; hop-timing jitter, bursty fades and interleaved emitters are future work. $M = 16$ confounds the full-state LCG (period ≤ 16) with a Markov cycle. The generators and experts share authors, which the held-out families mitigate but do not remove. The e-process guarantee is proven only for in-bank truths; out-of-bank protection is empirical, via H_0.

VII. CONCLUSION

OTS reduces a lossy passive sensor to one detection-level likelihood and reuses it across a heterogeneous generator bank, a training-free next-channel predictor, a change-point tracker and a conformal prediction set. The result is a system that names in-bank generators quickly and safely, keeps predicting

on generators it was never built for, re-identifies a mid-engagement change of algorithm, and attaches a distribution-free coverage guarantee to its forecast that transfers to the response. Each component has a published ancestor; the contribution is the sensor-aware assembly and its FH application. Future work targets mis-specified and bursty sensors from real captures, larger channel sets, and external learned baselines.

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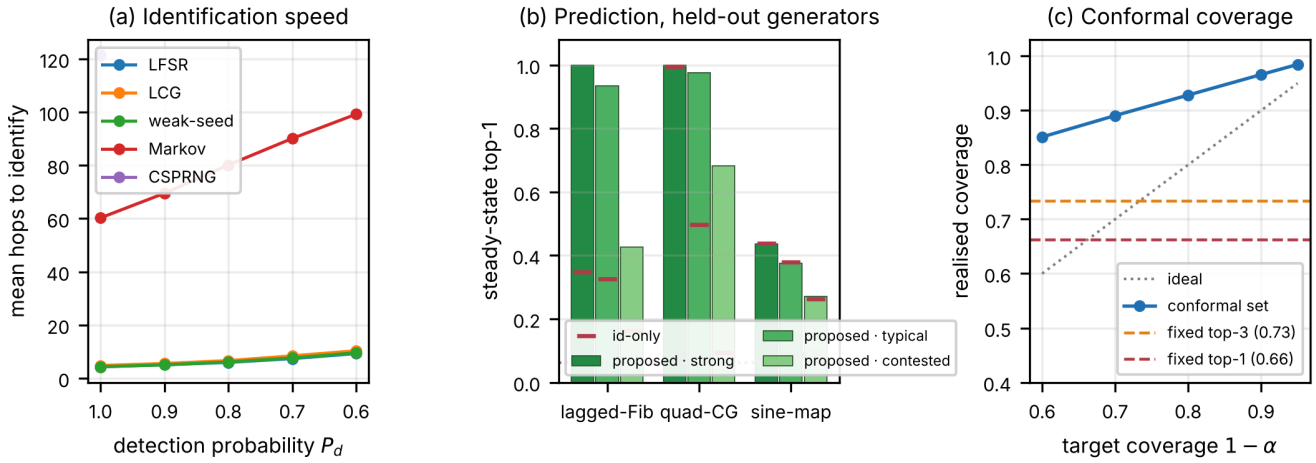


Fig. 5. (a) Mean hops-to-identify vs P_d: deterministic families need a handful of hops, Markov an order of magnitude more. (b) Steady-state top-1 next-channel accuracy on the three held-out generators (bars: proposed, across sensors; red ticks: identification-only bank); the dotted line is chance. (c) Conformal coverage calibration: the set tracks the target, fixed shortlists do not.